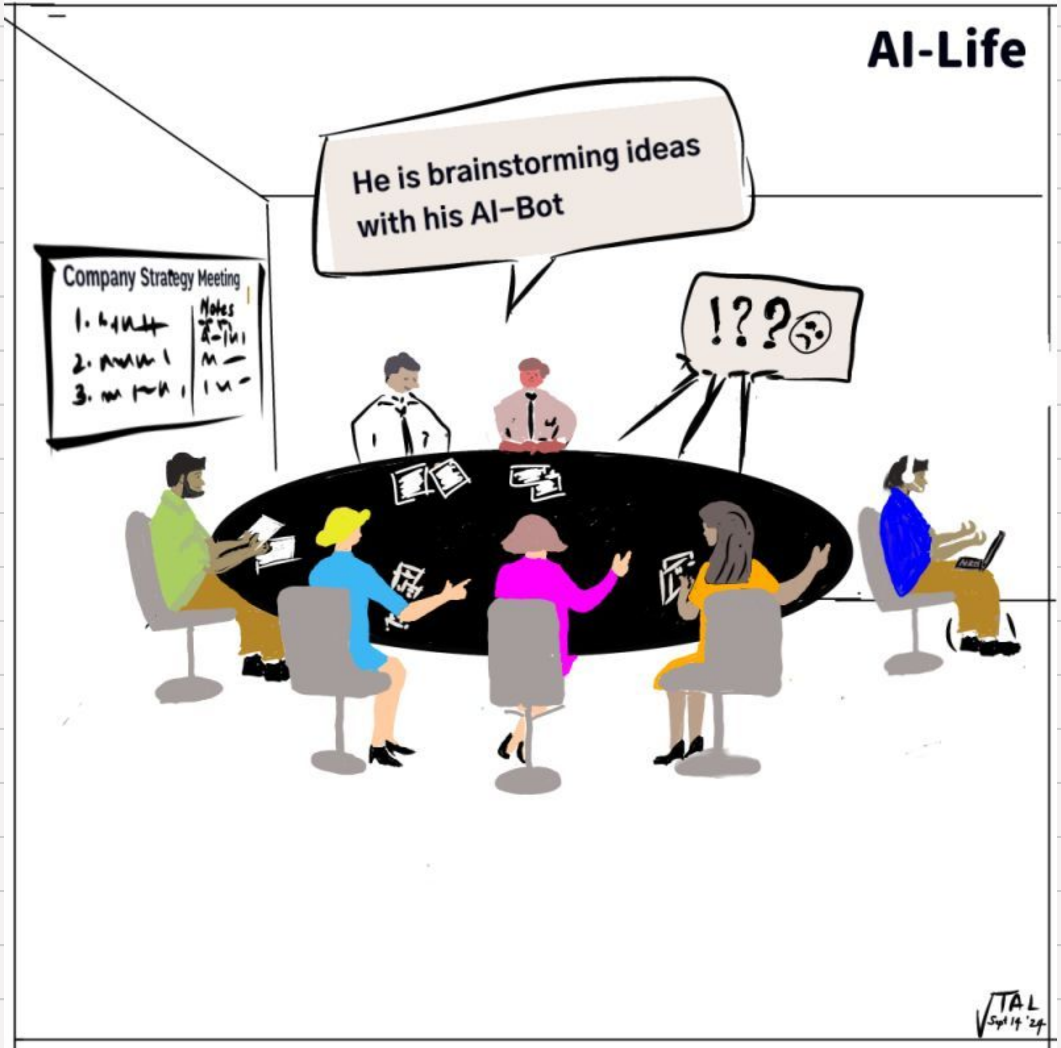

AI-Life
He is brainstorming ideas with his AI-Bot
Company Strategy Meeting
Notes
!??

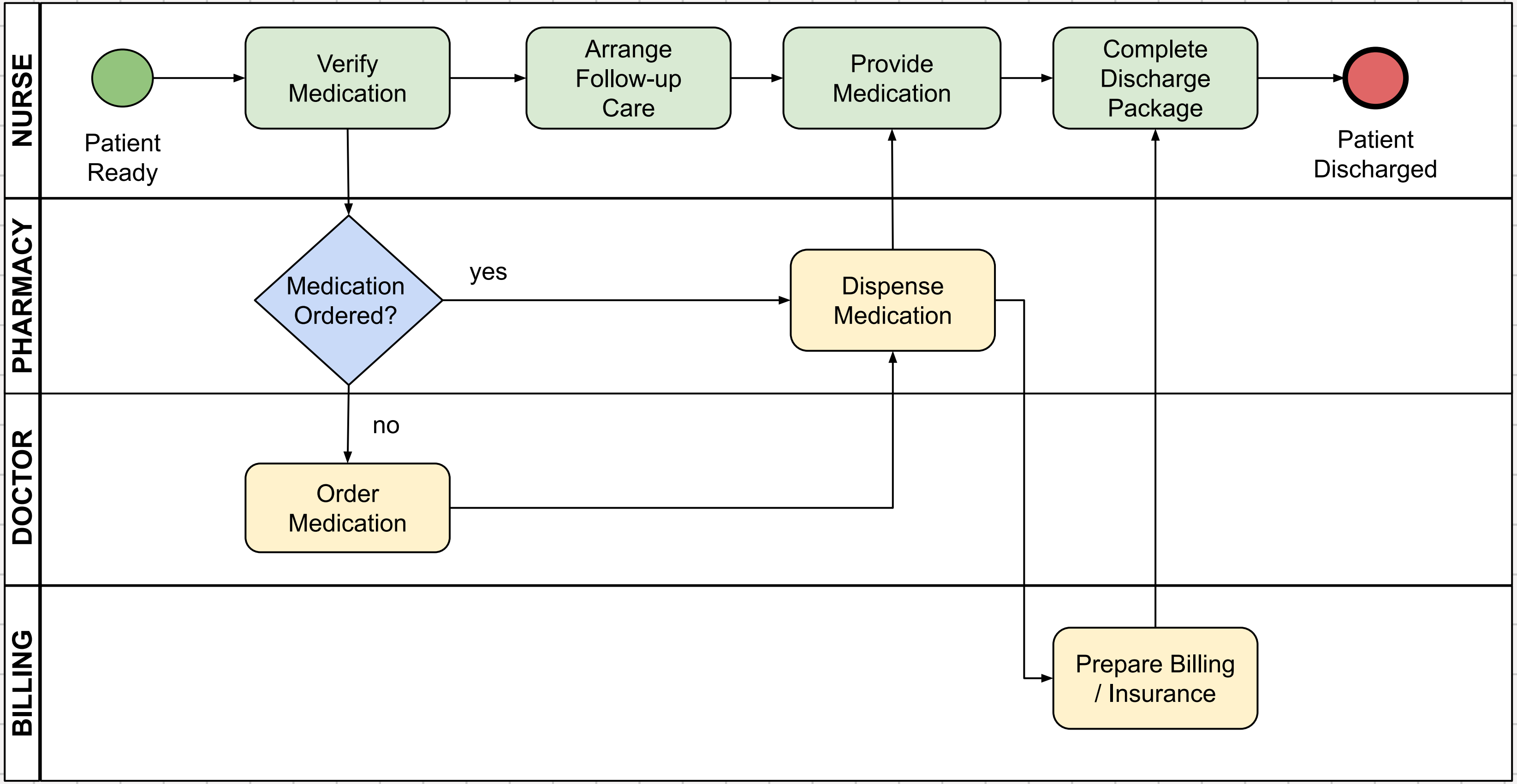
Announcements

NEWS

- Quiz #2 on Wed, April 1
  - Covers Lecture slides 7a, 7b, and Workshop Day 1
- Remainder of this semester will focus on business process automation using AI

Recap

Hospital Discharge Process



AI Agents

- “Discharge Coordination” Agent
- “Insurance Validation” Agent
- “Pharmacy Fulfillment” Agent
- “Billing” Agent
- “Communication” Agent

Results

- 20–30% productivity gain
- Discharge time reduced from 5 hrs → <1 hr
- SLA compliance ~98%, patient satisfaction +15–20%

Power of the Hybrid Approach

Capability	BPM Only	Agentic AI Only	Hybrid (BPM + Agentic AI)
Structure & Auditability	✅ High	⚠️ Low	✅ High
Adaptability to Edge Cases	⚠️ Low	✅ High	✅ High
Compliance & Governance	✅ High	⚠️ Medium	✅ High
Task Autonomy & Reasoning	❌ None	✅ Full	✅ Partial
Time to Value	⚠️ Medium	⚠️ Medium	✅ High (when modularized)
Learning & Optimization	❌ None	✅ Continuous	✅ Contextual Learning

Spring 2026

IS 365

Lecture 9:

Introduction to LLMs & Generative AI

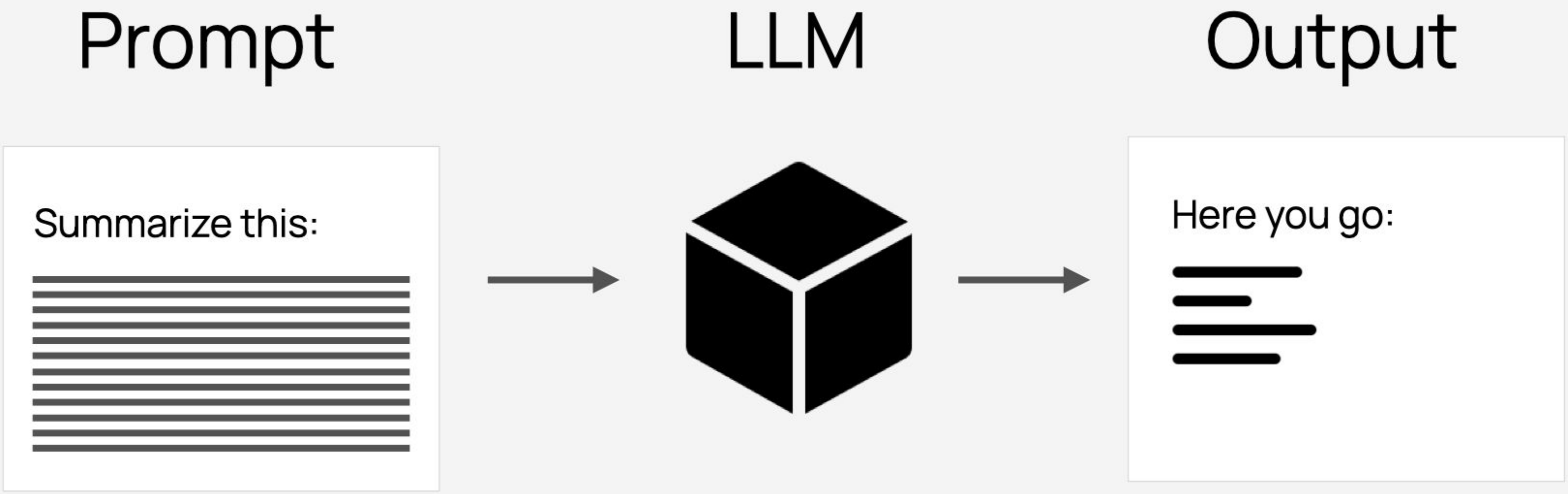
Early Conversational Agents

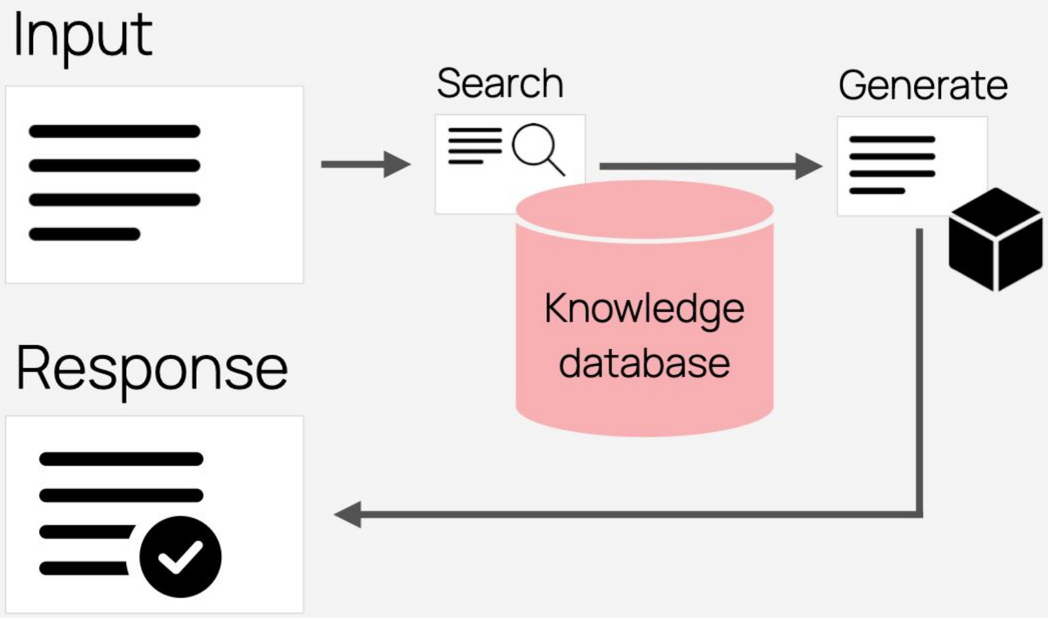
- Eliza (1966) and Parry (1972) marked the beginning of dialogue systems
- Early assistants like Siri, Alexa, and Cortana were limited in scope
- Relied on preconfigured intents and rule-based systems
- Laid the foundation for natural language human-machine interaction

ELIZA: Good Evening. Tell me your problems. Please terminate input with a period or a question mark.
PARRY: People get on my nerves sometimes.
ELIZA: I am not sure I understand you fully.
PARRY: You should pay more attention.
ELIZA: Suppose you should pay more attention.
PARRY: You're entitled to your own opinion.
ELIZA: What makes you think I am entitled to my own opinion?
PARRY: It bothers me just to be around people in general.
ELIZA: Please go on.
[Conversation continues...]
PARRY: I don't understand your motives.
ELIZA: Why don't you understand my motives?
PARRY: Shouldn't I be?
ELIZA: Why do you say that just now?
PARRY: That's just the way it is.
ELIZA: Suppose that's just the way it were not.
PARRY: So?
ELIZA: I am not sure I understand you fully.
PARRY: You harp on the same questions too much.
ELIZA: What does that similarity suggest to you?
[Conversation continues...]

1960s	Introducing Eliza, the first conversational agent
2011	Launch of modern conversational agents like Siri and Alexa
End of 2022	ChatGPT launches, marking the rise of generative AI
2024	Moving towards multimodal AI, integrati various input and output formats
February 2024	Evolution towards more autonomous and adaptable AI systems
January 2025	Introduction of highly autonomous AI agents such as Gemini 1.5, Deep Research, and Computer Use

Large Language Model (LLM)

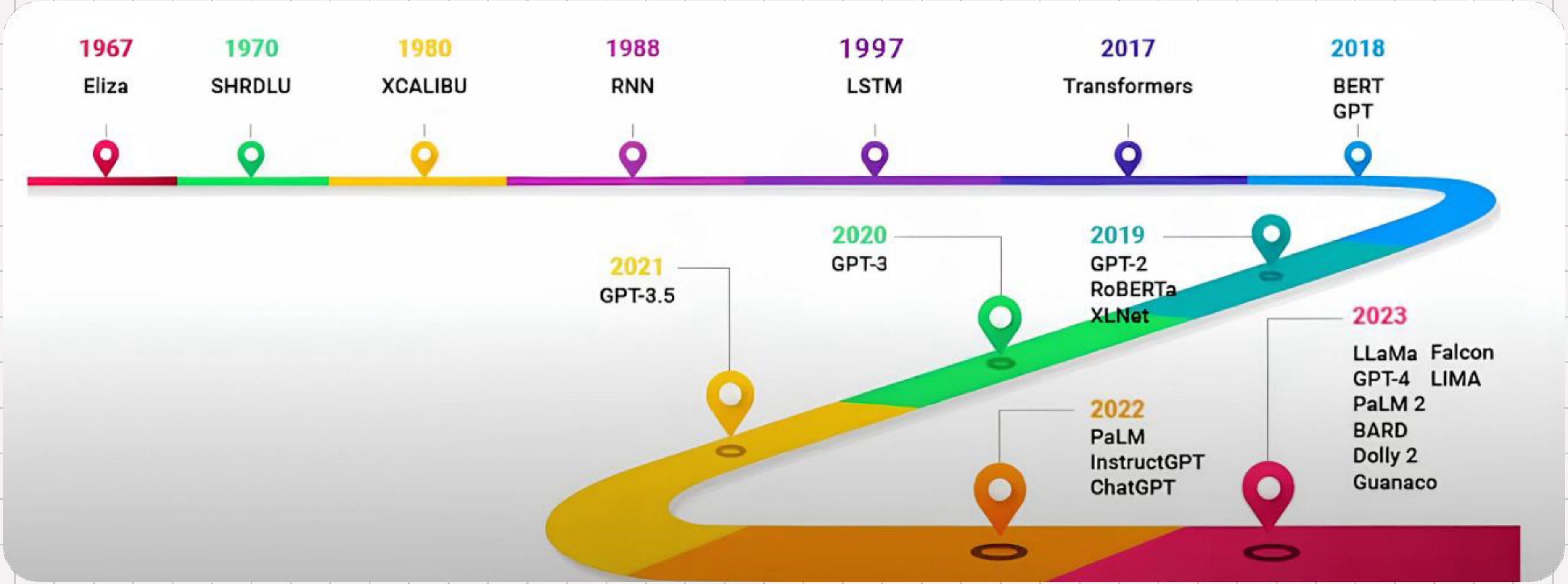

Prompt
LLM
Output
Summarize this:
Here you go:


Input
Search
Generate
Knowledge database
Response

A Large Language Models (LLM) is an advanced generative AI system capable of understanding and generating human-like text.

The model can recognize and interpret human languages, having been trained on millions of gigabytes of text data gathered from the internet.

Evolution of LLMs

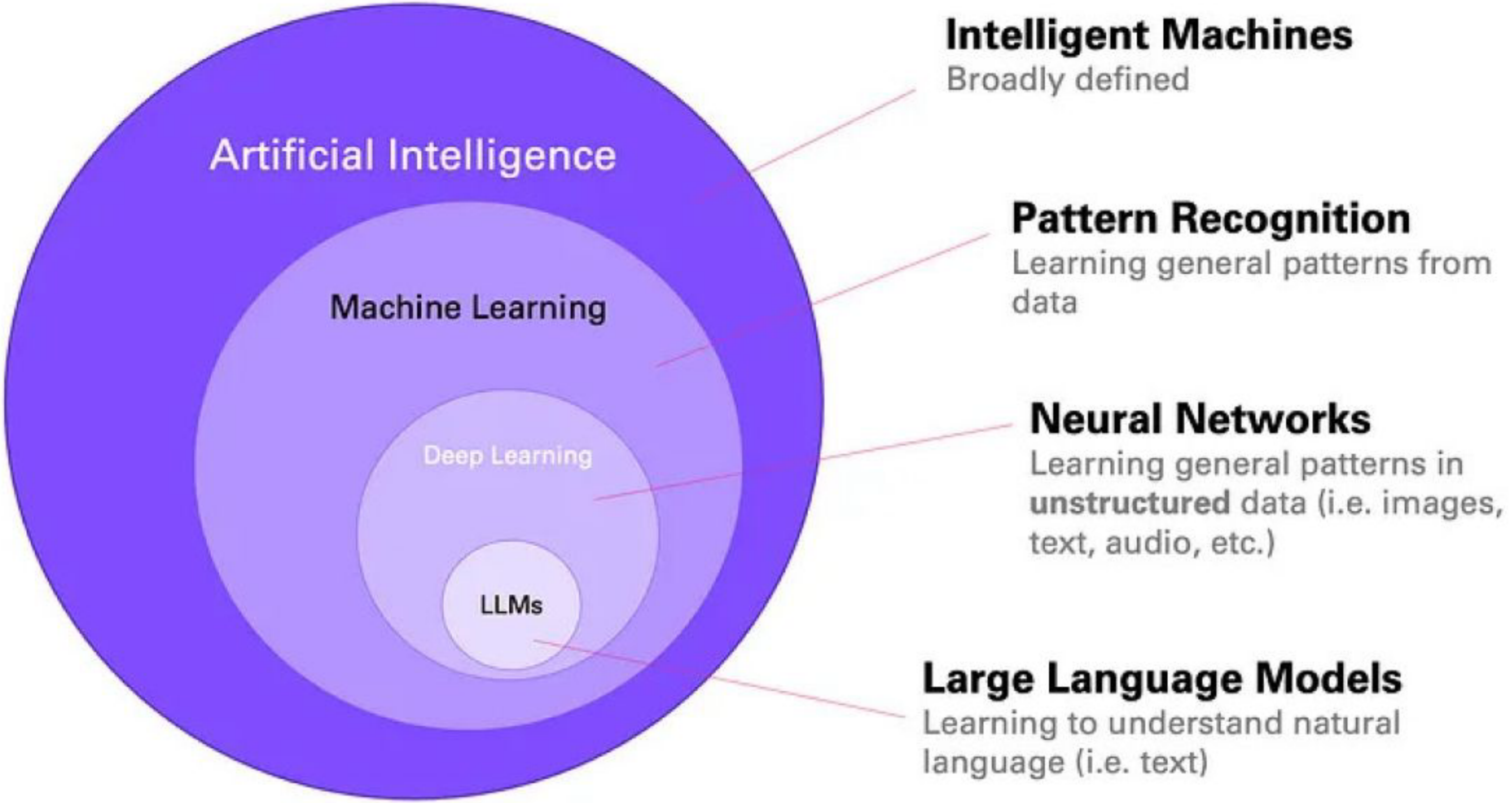


Capabilities

- Answers must be **correct:** match the source knowledge base.
- Answers must be **helpful**, **complete** and **relevant:** address the intent.
- Answer **tone** must be professional and match the company brand style.
- **Format:** ≤100 words, must include a link.

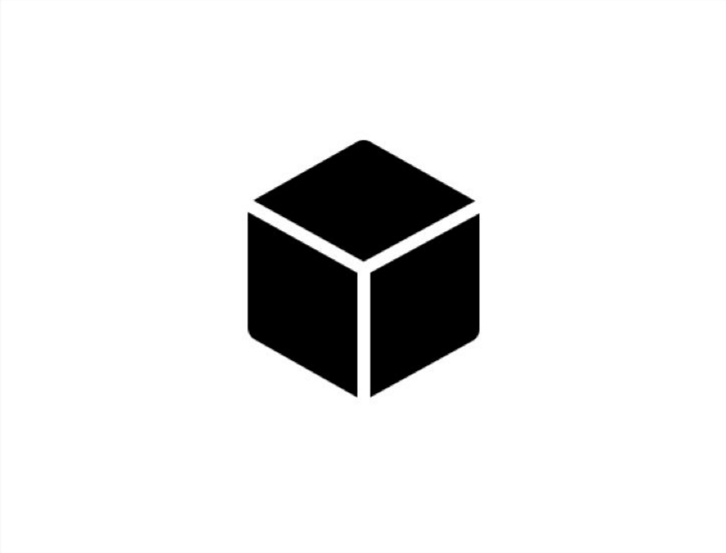
Risks

- **Hallucinations**: may give misleading answers.
- **Bias:** may propagate bias in its responses.
- **Toxicity:** may produce offensive outputs.
- **Sensitive data:** may expose private data.


Artificial Intelligence
Machine Learning
Deep Learning
LLMs
Intelligent Machines
Broadly defined
Pattern Recognition
Learning general patterns from data
Neural Networks
Learning general patterns in **unstructured** data (i.e. images, text, audio, etc.)
Large Language Models
Learning to understand natural language (i.e. text)

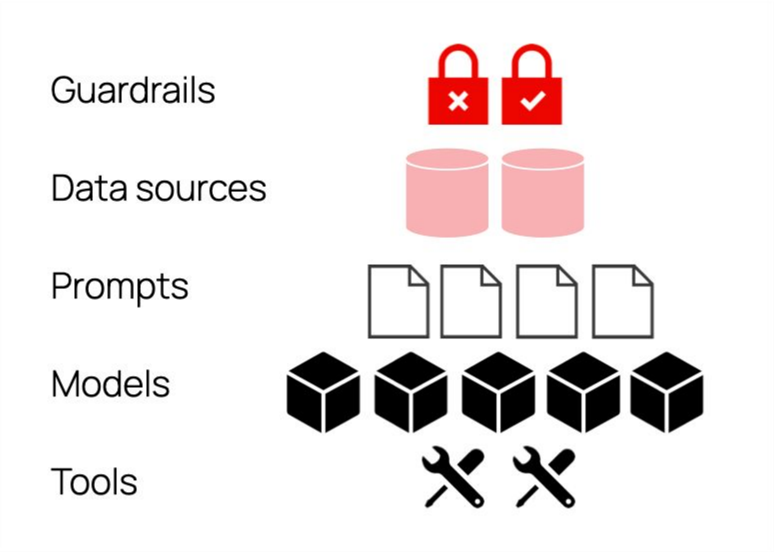
LLM Application: Legal

LLM

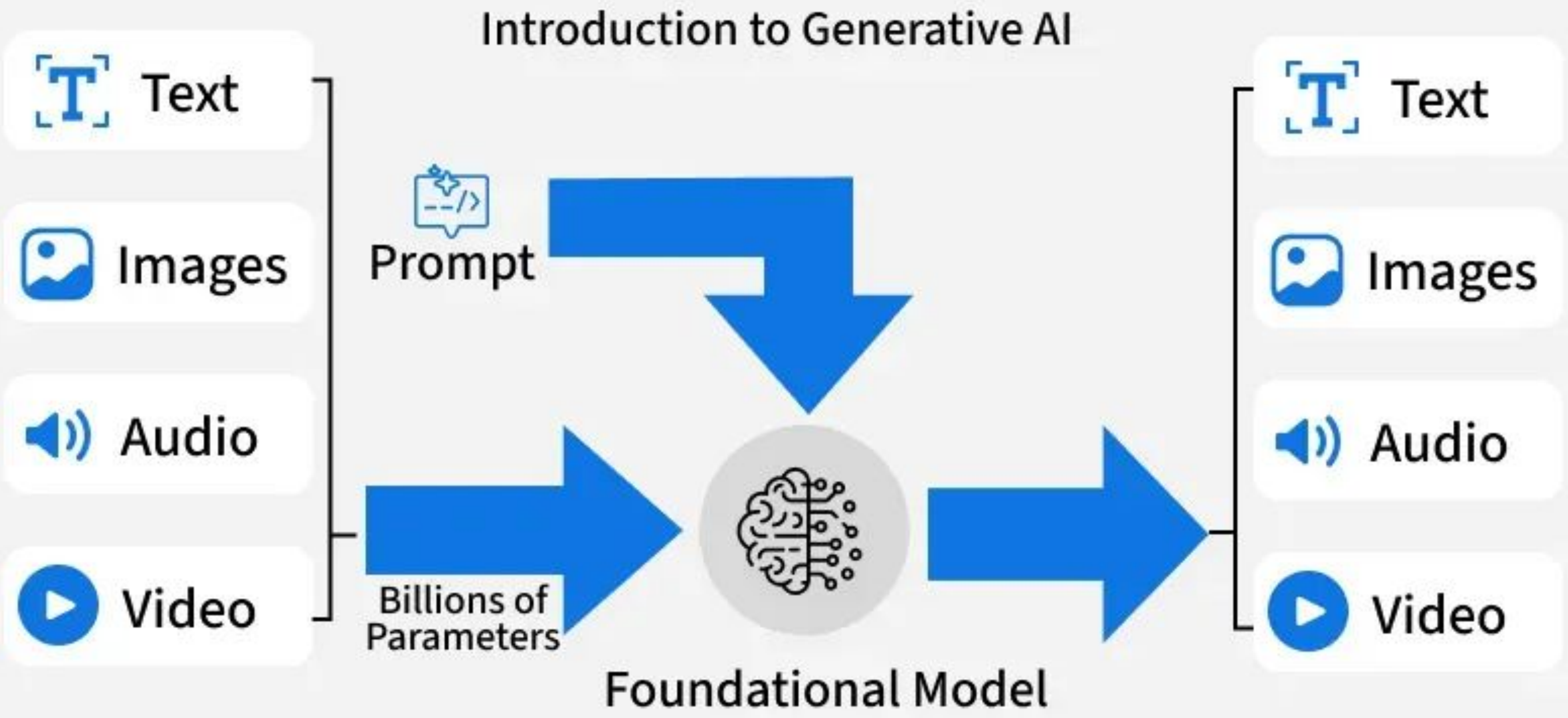


vs.

LLM-powered system



Generative AI

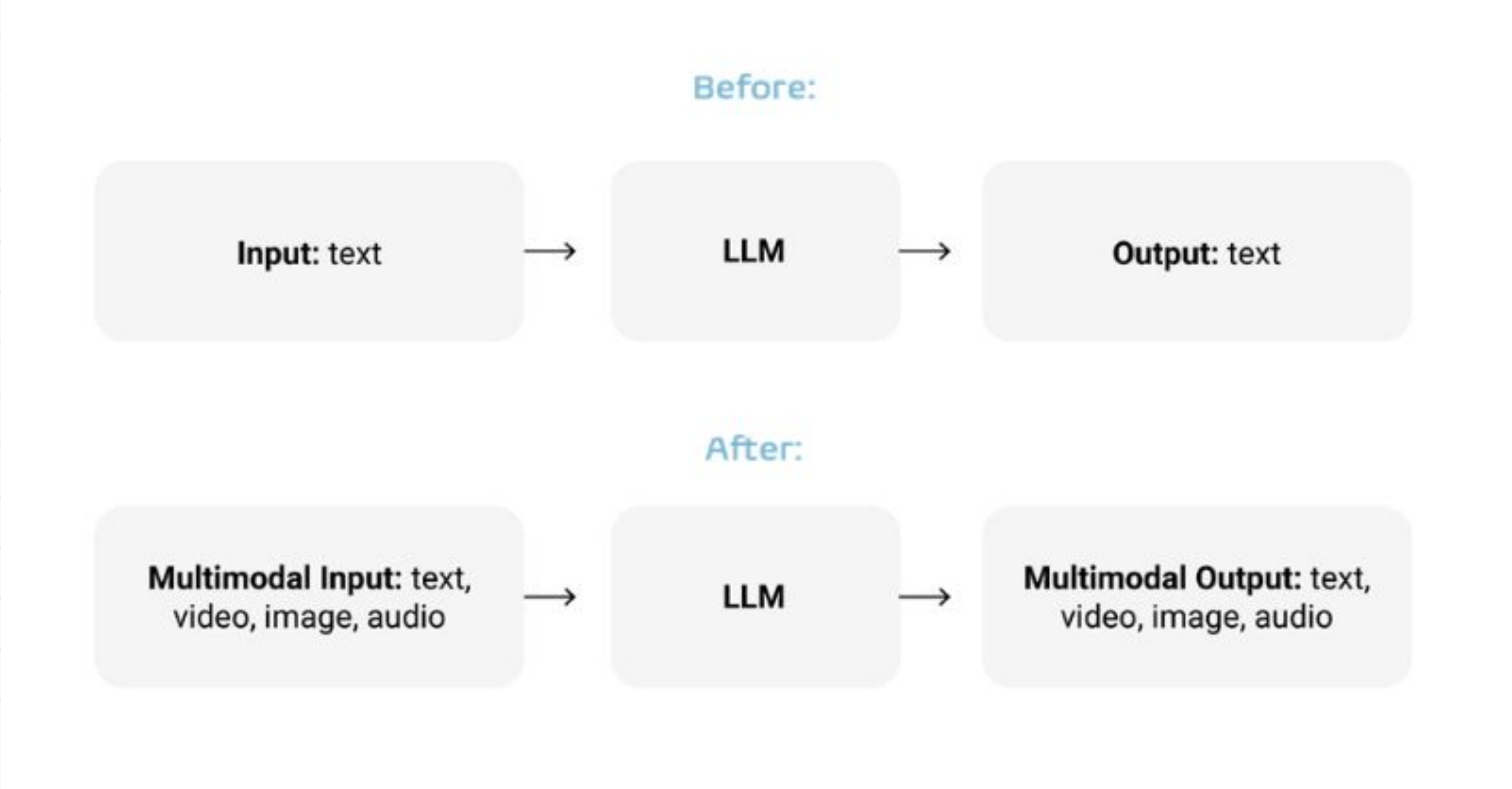

Introduction to Generative AI
Text
Images
Audio
Video
Prompt
Billions of Parameters
Foundational Model
Text
Images
Audio
Video

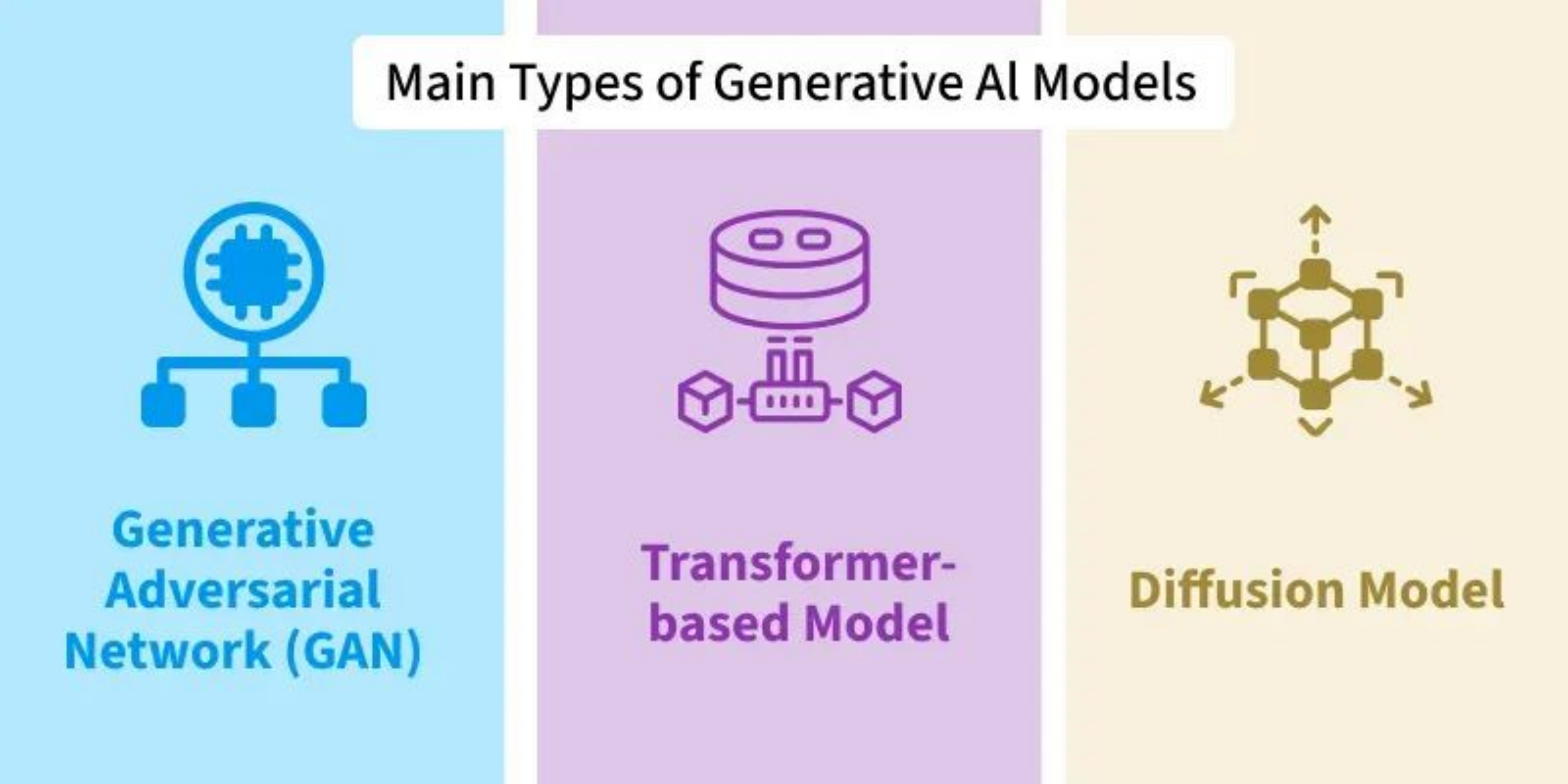
Rise of Generative AI

- Transformers and word prediction revolutionized natural language tasks
- ChatGPT (2022) introduced multitasking and multipurpose capabilities
- Democratized access to AI technology at scale
- Enabled autonomous task execution with minimal coding

The Multimodal Leap

- Integration of text, image, and audio processing in AI systems
- Generative models now handle text-to-image, video, and speech tasks
- Improves contextual understanding and user experience
- **Examples:** GPT-4+, Gemini, Sora, Flamingo, LLaMA 3+


Before:
Input: text
LLM
Output: text
After:
Multimodal Input: text, video, image, audio
LLM
Multimodal Output: text, video, image, audio


Main Types of Generative AI Models
Generative Adversarial Network (GAN)
Transformer-based Model
Diffusion Model

Generative Adversarial Network (GAN)

- GANs were among the first deep learning architectures for generative tasks
- A GAN uses two networks against each other – a generator that tries to produce realistic outputs and a discriminator that tries to detect fakes
- **Advantage:** Sharp, lifelike images though training
- **Limitation:** Can be unstable and prone to mode collapse

Transformer-based Model

- Autoregressive Transformer Models generate sequences by predicting the next token based on all previous ones moving step by step through the text
- **Advantage:** Fine grained control over each output step
- **Limitation:** Slower for long generations since tokens are generated one at a time
- **Example:** Generative Pre-trained Transformer (GPT) models which can produce coherent, context aware paragraphs, solve coding tasks or answer complex queries

Diffusion Model

- Diffusion models generate data (e.g., images, audio) by starting with pure random noise and gradually refining through a series of denoising steps
- Used for inpainting, style transfer, and conditional generation from text prompts
- **Advantage:** Highly detailed and realistic results specially in image synthesis
- **Examples:** Stable Diffusion, DALL·E 3

Primary AI Use Cases

- Content generation from long-form writing to marketing copy
- Coding and support for developer workflows
- Graphics and image generation
- Video generation
- Predictive analytics, structured data tasks, and reasoning
- Voice / multimodal input handling

#	MODEL	LAUNCH	PRIMARY USE CASES
1	**GPT-4o** OpenAI	May 2024	Content generation, Coding / copilot, Multimodal reasoning, Voice interaction
2	**Claude 3.7 Sonnet** Anthropic	Feb 2025	Content generation, Coding / copilot, Extended thinking, Analysis & research
3	**Gemini 2.0 Flash** Google DeepMind	Feb 2025	Search augmentation, Multimodal input, Coding assistance, Agentic workflows
4	**Midjourney v6.1** Midjourney	Jul 2024	Image generation, Creative design, Marketing visuals
5	**DALL·E 3** OpenAI	Oct 2023	Image generation, Illustration, Text-to-image

#	MODEL	LAUNCH	PRIMARY USE CASES
6	**GitHub Copilot (GPT-4o)** Microsoft / OpenAI	2024	Code completion, Dev copiloting, Test generation
7	**Sora** OpenAI	Feb 2024	Video generation, Visual storytelling, Creative media
8	**DeepSeek R1** DeepSeek	Jan 2025	Predictive analytics, Coding & math, Chain-of-thought reasoning
9	**Llama 3.1 405B** Meta	Jul 2024	Content generation, Enterprise analytics, Open-source fine-tuning
10	**Stable Diffusion 3.5** Stability AI	Oct 2024	Graphics generation, Product design, Open-weight deployment